THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For a force to act, there must be at least: 1

(a) one object **(b)** two interacting objects
(c) three objects **(d)** no objects

2. If an object is at rest, the forces acting on it are: 1

(a) only magnetic **(b)** unbalanced **(c)** absent always **(d)** balanced

3. Friction arises because of: 1

(a) smoothness of air **(b)** irregularities of surfaces in contact
(c) gravity only **(d)** magnetism

4. Like poles of magnets: 1

(a) do nothing **(b)** melt **(c)** repel **(d)** attract

5. The two kinds of static charge are: 1

(a) north and south **(b)** positive and negative
(c) high and low **(d)** hot and cold

6. A force of 240 N acts perpendicularly on an area of 4 m². The pressure is: 1

(a) 960 N/m² **(b)** 120 N/m² **(c)** 60 N/m² **(d)** 30 N/m²

7. A nail is driven easily by its pointed end because: 1

(a) the pressure is zero **(b)** the force is zero
(c) the area is large **(d)** the area is small so pressure is large

8. Overhead water tanks are placed at a height to: 1

(a) make water warm **(b)** increase water pressure at taps
(c) reduce pressure **(d)** store more heat

9. We are not crushed by atmospheric pressure because: 1

(a) air is very light **(b)** we are strong
(c) pressure inside our body balances it **(d)** air has no weight

10. Faster wind speed means: 1

(a) higher air pressure **(b)** no change
(c) zero air **(d)** lower air pressure

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does a ball thrown upward come back? 2

12. Distinguish between mass and weight. 2

13. State the key principle for the formation of wind. 2

14. What is a sea breeze? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the activity that shows upthrust using an empty bottle. 3

16. Explain the formation of a cyclone. 3

17. State the condition for sinking and floating using gravitational force and buoyant force. 3

OR

Explain why pumice floats on water. 3

18. Name the structure of a cyclone and describe the eye. 3

OR

List three destructive effects of cyclones. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. A stone of mass 2 kg is on Earth, Moon and Mars. Find weights and explain why they differ. (Earth 10, Moon 1.6, Mars 3.8 N/kg) 5

OR

Describe vertical motion under gravity and explain the changes in speed while going up and coming down. 5

20. Describe the formation of a cyclone and its structure. 5

OR

Describe the destructive effects of cyclones and the precautions that can reduce loss. 5

END OF QUESTION PAPER — Total: 40 Marks

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